

## **Software Design Document**

### **Secure Deep Scan**



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## Summary

The Secure Deep Scan System is a web based platform designed to scan websites for common web application vulnerabilities and security attacks using the OWASP ZAP API. The system helps users identify vulnerabilities such as SQL Injection (SQLi), Cross-Site Scripting (XSS), broken authentication, security misconfigurations, and other OWASP Top 10 vulnerabilities through automated website scanning and report generation. The system allows users to submit website URLs for security scanning, perform automated vulnerability analysis, view detailed scan reports, and monitor scan history through a secure dashboard. The application follows MVC architecture using Django and MySQL to ensure scalability, maintainability, and secure data handling. The project purpose to provide an efficient, intelligent, and user-friendly cybersecurity solution for secure against digital growing attacks on Internet.

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# 1. Application Architecture

In This project , I am using modern architecture of website development which is MVC. It is widely used and currently standard format to build websites. Which can be easily scale and maintained.

## 1.1 MVC Architecture

| Component  | Purpose                               |
|------------|---------------------------------------|
| Model      | Handles database operations           |
| View       | Displays graphical user interface     |
| Controller | Processes requests and business logic |

## 1.2 Proposed Models

- User
- WebsiteScan
- ScanReport
- ThreatDetection

## 1.3 Proposed Views

### Views

- Dashboard View
- Website Scan View
- Scan Report View

## 1.4 Proposed Controllers

Proposed Controller are controllers which will be used in developing website. Basically, A controller plays an important role in perfect working of website with efficiency. A Controller is Class where we define main business logic of website. In this Section, I will define expected controllers which will be used in my website.

#### 1.4.1 Website Scan Controller

Functions:

- submitURL()
- startScan()
- generateReport()
- detectVulnerabilities()

#### 1.4.2 Notification Controller

Functions:

- sendAlert()
- sendReportNotification()

### 1.5 APIs and External Libraries

| API/Library     | Purpose                                                                                                                                                               |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Django          | It is Backend framework of project. It provides all basic code structure. I just need to focus on building logical unit for my website.                               |
| MySQL Connector | Help in connecting with Database                                                                                                                                      |
| OWASP ZAP API   | It is main thing which provide main functionality of website. It provide vulnerability scanning engine to scan and generate insightful raw information about website. |
| SMTP Service    | Email notifications                                                                                                                                                   |

### 1.6 UML Diagrams

The following diagrams explain the working of the system:

- Use Case Diagram
- Sequence Diagram
- ER Diagram
- Data Flow Diagram

These diagrams are mentioned in my SRS with explanation.

Here is reference to study more about my diagrams.

([Reference to Diagrams](#)).

## 1.7 Coding Conventions

| Element     | Convention |
|-------------|------------|
| Class Names | PascalCase |
| Variables   | snake_case |
| Functions   | snake_case |
| Constants   | UPPER_CASE |

## 2. Data Model Schema

The Secure Deep Scan System uses a relational database schema to manage users, website scans, vulnerability reports, detected threats.

### 2.1 Entities

Entities are the table in Database. These act as a blueprint for data. These entities tell what and which data to store in what which. These define rule, laws & data type for data. Following are entities which will be used in my project database.

### 2.1.1 Website Scan Entity

| Attribute   | Description        |
|-------------|--------------------|
| scan_id     | Unique scan ID     |
| website_url | Target website URL |
| scan_date   | Scan timestamp     |

### 2.1.2 Scan Report Entity

| Attribute      | Description         |
|----------------|---------------------|
| report_id      | Unique report ID    |
| scan_date      | Scan timestamp      |
| risk_level     | Threat severity     |
| result_summary | Scan summary        |
| scan_id        | Linked website scan |

### 2.1.3 Threat Detection Entity

| Attribute   | Description      |
|-------------|------------------|
| threat_id   | Unique threat ID |
| threat_name | Threat name      |
| threat_type | Malware/Virus    |
| severity    | Threat severity  |
| report_id   | Linked report    |

### 2.1.4 System Log Entity

| Attribute | Description            |
|-----------|------------------------|
| log_id    | Unique log ID          |
| activity  | Activity description   |
| log_time  | Timestamp              |
| status    | Success/Failed         |
| user_id   | Linked user IP address |

### 2.1.5 Entity Relationships

| Relationship                   | Type        |
|--------------------------------|-------------|
| User → Uploaded File           | One-to-Many |
| Uploaded File → Scan Report    | One-to-Many |
| Scan Report → Threat Detection | One-to-Many |
| User → System Log              | One-to-Many |

## 2.2 Database Constraints

The database includes constraints to maintain data integrity.

Constraints Used in my project database are given as below:-

| Constraint  | Purpose                  |
|-------------|--------------------------|
| Primary Key | Unique identification    |
| Foreign Key | Maintain relationships   |
| NOT NULL    | Prevent empty fields     |
| UNIQUE      | Prevent duplicate emails |

| Constraint | Purpose              |
|------------|----------------------|
| CHECK      | Validate data values |

## 2.3 Future Database Enhancements

Future versions of the database may include:

- Admin Dashboard
- User Profile Managemnet
- History specific to each User

## 3. User Interface

User Interface is the actual visual interface/page. User need visual UI to work and understand functionality provide by website. That's why today UI is necessary part of software/Web development. For my project, I will use Blue Royal Navy themed, Easy to understand UI. So , User can easily interact with website.

### 3.1 User Interface Design Philosophy

This project follows a modern, clean, and responsive user interface design to provide with smooth experience for users. The interface is designed according to Human Computer Interaction to ensure simplicity, accessibility, readability, and ease of navigation.

The overall UI focuses on:

- Minimal and modern design
- Easy & Understandable pages
- Only desktop responsiveness
- Fast interaction

### 3.2 Color Scheme

The system uses an Royal-Navy blue color palette to represent sustainability and cybersecurity/Terminal related themed.

| Purpose         | Color                |
|-----------------|----------------------|
| Primary Color   | Dark Royal Navy Blue |
| Secondary Color | Dark Light Navy Blue |
| Error Messages  | Red                  |

### 3.3 Typography

The system will uses simple and readable fonts for better accessibility.

| Element   | Font Style                   |
|-----------|------------------------------|
| Headings  | Bold Times new roman         |
| Body Text | Regular times new roman      |
| Buttons   | Medium-weight time new roman |

#### Font Features

- Clear readability
- Professional appearance

### 3.4 Responsiveness and Screen Constraints

The system is fully responsive and supports multiple screen sizes.

### **Supported Devices**

- Laptops
- Desktop

### **Responsive Features**

- Flexible layouts
- Adaptive navigation menus

### **Recommended Resolution**

- $1920 \times 1080$  for desktop

## **3.5 User Interface Screens**

### **3.5.1 Home Page**

This is Home page. when user search website and click URL of website, this is first page he/she see. This page provide all necessary information user need.

#### **Interaction :-**

User enter URL of his/her website and press launch scan button and rest of work done by web engine to find all vulnerability of entered website.

#### **Pre Condition :**

Users must have internet.

Users must know how to copy and paste correct URL

Users have basic knowledge of using Modern Website.

Home Page View of With input field to input URL and launch button to start scanning

**GuardScan**

New Scan

History

Reports

Settings

## Security Assessment

Scan any web application for logic loopholes and backdoors.

Target URL

https://example-target.com

Scan Depth

Full Audit

Launch Scan

**SCANNING PROGRESS**

**75%**

**HIGH RISKS**

**12**

Critical loopholes detected

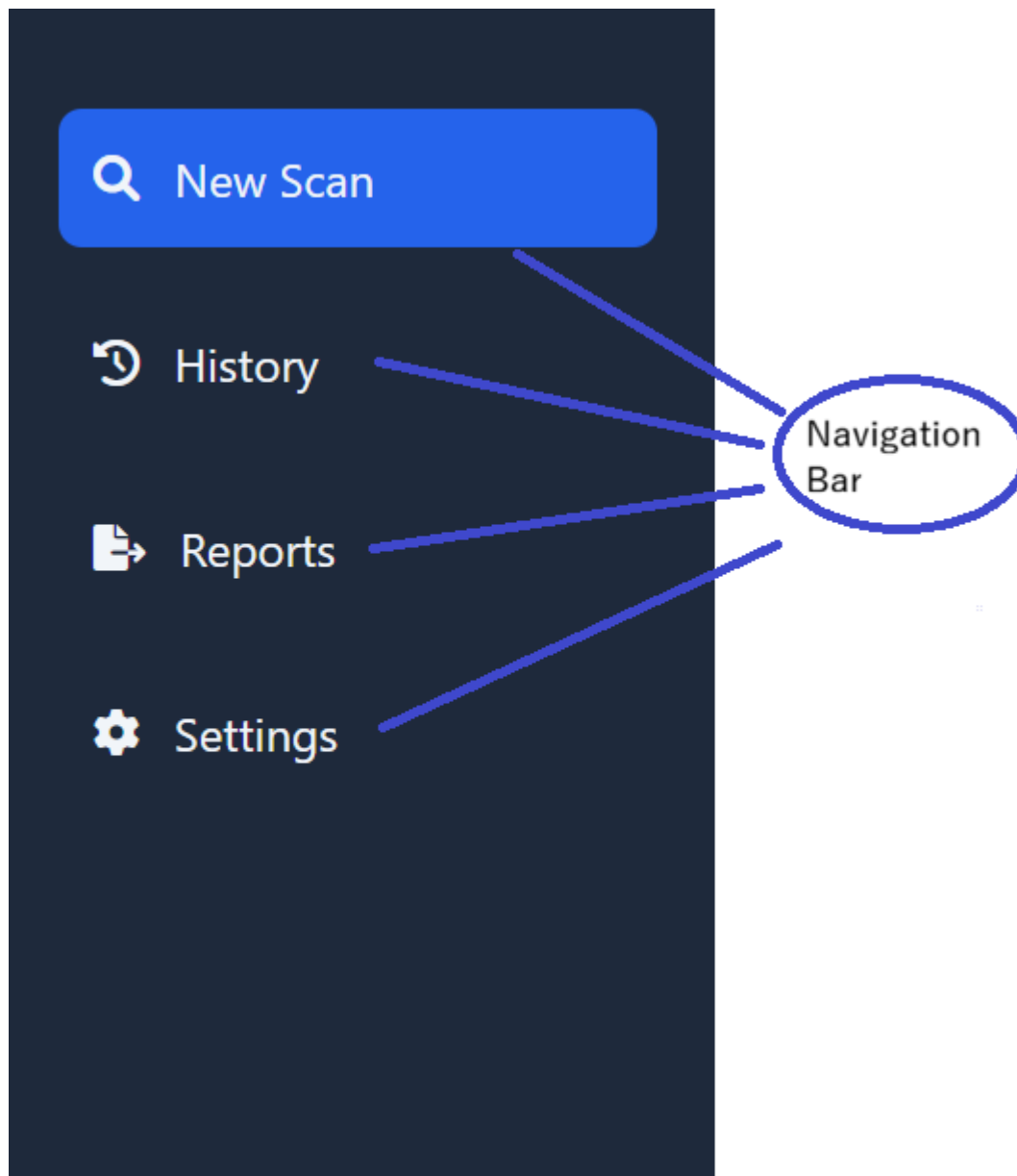
**IP ANALYZED**

**192.168.1.104**

External Black-box Mode

### 3.5.2 Navigation Page

It is part of home page. It provide basic navigation to other pages.



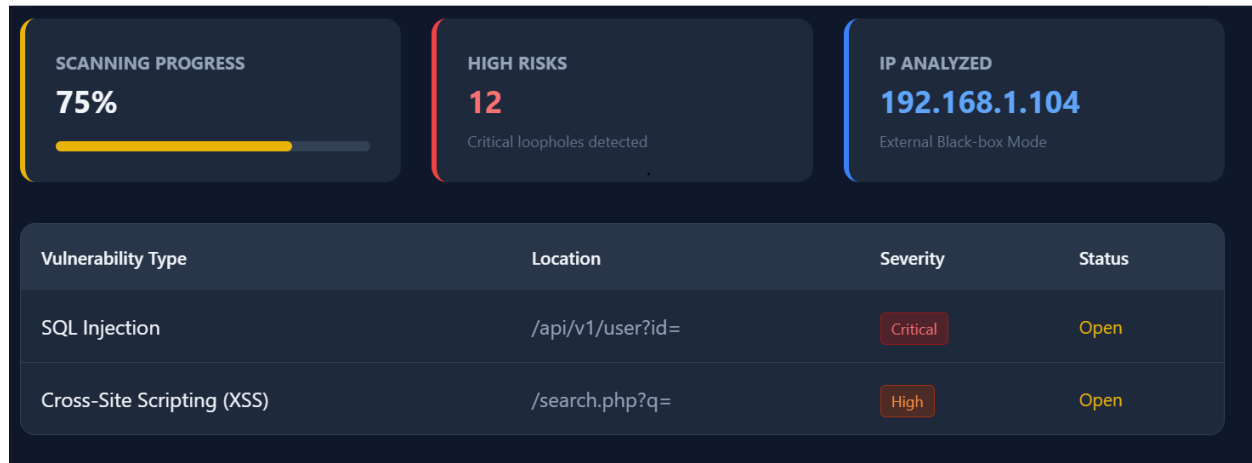
### 3.5.3 Report View Page

It is main part where result of target website will be uploaded. This page provide threats , vulnerabilities and their solution in a visual listing format.

**Pre-Condition:**

Target web scanning run successfully.

Example View of Report Data after successful scan of Website



### 3.9 Future UI Enhancements

Future versions may include:

- Dark mode (Black jet and silver-black Color theme)
- Voice assistance
- Mobile application interface
- AI-based personalized dashboard

These are not immediately added after release but gradually update in multiple versions.

## 4. References

- [OWASP Foundation, “OWASP Top 10:2021,” 2021](#)
- [OWASP Foundation, “OWASP Web Security Testing Guide v4.2,” 2020](#)

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- [D. Stuttard and M. Pinto, The Web Application Hacker's Handbook: Finding and Exploiting Security Flaws, 2nd ed. Indianapolis, IN, USA: Wiley, 2011.\[5\] FIRST.Org, Inc., "Common Vulnerability Scoring System version 3.1: Specification Document," 2019](#)
- [L. Richardson, "Beautiful Soup Documentation," 2024](#)